WSEI University in Lublin

Faculty of Transportation and Information

Virtualization techniques, containerization

Laboratory

Kubernetes

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Lab 4 – Kubernetes

1. **Which types of applications are best suited for a microservices architecture?**
   Applications that need to grow easily, remain reliable under failure, and adapt quickly benefit most from microservices. Examples include online shopping systems, video streaming platforms, and social networking applications.

2. **Give an example of how a computing cluster can be used.**
   Computing clusters are commonly used in big data environments such as Apache Hadoop, where several machines cooperate to analyze and process massive datasets at the same time.

3. **What are the key benefits of using Kubernetes?**
   - Automatic deployment and scaling of applications
   - Ability to recover from failures automatically
   - Built-in load distribution
   - Efficient use of system resources
   - Support for smooth updates and easy rollbacks

4. **What is the commonly used short form of Kubernetes?**
   - K8s

5. **What are the two essential parts of a Kubernetes cluster?**
   - The Control Plane
   - Worker Nodes

6. **What job roles could you qualify for with Docker and Kubernetes skills?**
   - Kubernetes Administrator
   - DevOps Engineer
   - Cloud Engineer
   - Site Reliability Engineer (SRE)
   - Software Engineer
   -

Exercise : Exploring Kubernetes Cluster functionality

Installing latest version of multipass from

https://github.com/canonical/multipass/releases

```
PS C:\WINDOWS\system32> multipass launch --name microk8s-vm --memory 4G --disk 40G
>>
Launched: microk8s-vm
PS C:\WINDOWS\system32> X
```

multipass shell microk8s-vm command to access the virtual machine instance.

```
Launched: microk8s-vm
PS C:\WINDOWS\system32> multipass shell microk8s-vm
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-51-generic x86_64)
```

Install the MicroK8s tool snap-in application

```
ubuntu@microk8s-vm:~$ sudo snap install microk8s --classic
2025-02-07T13:51:04Z INFO Waiting for automatic snapd restart...
microk8s (1.31/stable) v1.31.5 from Canonical✓ installed
ubuntu@microk8s-vm:~$
```

Checking the status of the installation

```
ubuntu@microk8s-vm:~$ sudo microk8s status --wait-ready
microk8s is running
high-availability: no
  datastore master nodes: 127.0.0.1:19001
  datastore standby nodes: none
addons:
  enabled:
    dns                  # (core) CoreDNS
    ha-cluster           # (core) Configure high availability on the current node
    helm                 # (core) Helm - the package manager for Kubernetes
    helm3                # (core) Helm 3 - the package manager for Kubernetes
  disabled:
    cert-manager         # (core) Cloud native certificate management
    cis-hardening        # (core) Apply CIS K8s hardening
    community            # (core) The community addons repository
    dashboard            # (core) The Kubernetes dashboard
    gpu                  # (core) Alias to nvidia add-on
    host-access          # (core) Allow Pods connecting to Host services smoothly
    hostpath-storage     # (core) Storage class; allocates storage from host directory
    ingress              # (core) Ingress controller for external access
    kube-ovn             # (core) An advanced network fabric for Kubernetes
    mayastor             # (core) OpenEBS MayaStor
    metallb              # (core) Loadbalancer for your Kubernetes cluster
    metrics-server       # (core) K8s Metrics Server for API access to service metrics
    minio                # (core) MinIO object storage
    nvidia               # (core) NVIDIA hardware (GPU and network) support
    observability        # (core) A lightweight observability stack for logs, traces and metrics
    prometheus           # (core) Prometheus operator for monitoring and logging
    rbac                 # (core) Role-Based Access Control for authorisation
    registry             # (core) Private image registry exposed on localhost:32000
    rook-ceph            # (core) Distributed Ceph storage using Rook
    storage              # (core) Alias to hostpath-storage add-on, deprecated
ubuntu@microk8s-vm:~$
```

installing addons

```
deployment.apps/hostpath-provisioner created
storageclass.storage.k8s.io/microk8s-hostpath created
serviceaccount/microk8s-hostpath created
clusterrole.rbac.authorization.k8s.io/microk8s-hostpath created
clusterrolebinding.rbac.authorization.k8s.io/microk8s-hostpath created
Storage will be available soon.
The registry will be created with the size of 20Gi.
Default storage class will be used.
namespace/container-registry created
persistentvolumeclaim/registry-claim created
deployment.apps/registry created
service/registry created
configmap/local-registry-hosting configured
ubuntu@microk8s-vm:~$
```

Creating an alias for microk8s.kubectl

```
ubuntu@microk8s-vm:~$ sudo snap alias microk8s.kubectl kubectl
Added:
  - microk8s.kubectl as kubectl
ubuntu@microk8s-vm:~$
```

Viewing cluster node information

```
ubuntu@microk8s-vm:~$ sudo kubectl get nodes
NAME          STATUS   ROLES    AGE   VERSION
microk8s-vm   Ready    <none>   15m   v1.31.5
ubuntu@microk8s-vm:~$
```

Additional information from the API server

```
ubuntu@microk8s-vm:~$ sudo kubectl get services -o wide
NAME         TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)   AGE   SELECTOR
kubernetes   ClusterIP   10.152.183.1   <none>        443/TCP   64m   <none>
ubuntu@microk8s-vm:~$
```

All services in all the namespaces

```
ubuntu@microk8s-vm:~$ sudo kubectl get services -o wide --all-namespaces
NAMESPACE            NAME                        TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)
                AGE   SELECTOR
container-registry   registry                    NodePort    10.152.183.233   <none>        5000:3200
0/TCP           14m   app=registry
default              kubernetes                  ClusterIP   10.152.183.1     <none>        443/TCP
                23m   <none>
kube-system          dashboard-metrics-scraper   ClusterIP   10.152.183.135   <none>        8000/TCP
                14m   k8s-app=dashboard-metrics-scraper
kube-system          kube-dns                    ClusterIP   10.152.183.10    <none>        53/UDP,53
/TCP,9153/TCP   23m   k8s-app=kube-dns
kube-system          kubernetes-dashboard        ClusterIP   10.152.183.97    <none>        443/TCP
                14m   k8s-app=kubernetes-dashboard
kube-system          metrics-server              ClusterIP   10.152.183.87    <none>        443/TCP
                14m   k8s-app=metrics-server
ubuntu@microk8s-vm:~$
```

Creating an NGINX server deployment

```
ubuntu@microk8s-vm:~$ sudo kubectl create deployment nginx --image=nginx
deployment.apps/nginx created
ubuntu@microk8s-vm:~$
```

To retrieve information about the cluster pods

```
ubuntu@microk8s-vm:~$ sudo kubectl get pods
NAME                      READY   STATUS    RESTARTS   AGE
nginx-676b6c5bbc-v98jh    1/1     Running   0          2m29s
ubuntu@microk8s-vm:~$
```

To find the pod address pass the-o wide parameter

```
ubuntu@microk8s-vm:~$ sudo kubectl get pods -o wide
NAME                      READY   STATUS    RESTARTS   AGE     IP           NODE         NOMINATED NODE   READINESS GATES
nginx-676b6c5bbc-v98jh    1/1     Running   0          3m31s   10.1.254.73  microk8s-vm  <none>           <none>
ubuntu@microk8s-vm:~$
```

Accesing the website

```
ubuntu@microk8s-vm:~$ wget 10.1.254.73
--2025-02-07 14:35:48--  http://10.1.254.73/
Connecting to 10.1.254.73:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 615 [text/html]
Saving to: ‘index.html’

index.html                100%[===================================>]     615  --.-KB/s    in 0.008s

2025-02-07 14:35:49 (71.5 KB/s) - ‘index.html’ saved [615/615]

ubuntu@microk8s-vm:~$
```

Increase the total number of NGINX pods to three

```
ubuntu@microk8s-vm:~$ sudo kubectl scale --replicas=3 deployments/nginx
deployment.apps/nginx scaled
ubuntu@microk8s-vm:~$
```

checking the number of pods running

```
ubuntu@microk8s-vm:~$ sudo kubectl get pods -o wide
NAME                      READY   STATUS    RESTARTS   AGE   IP           NODE         NOMINATED NODE   READINESS GATES
nginx-676b6c5bbc-27djh    1/1     Running   0          43s   10.1.254.75  microk8s-vm  <none>           <none>
nginx-676b6c5bbc-pm57x    1/1     Running   0          43s   10.1.254.74  microk8s-vm  <none>           <none>
nginx-676b6c5bbc-v98jh    1/1     Running   0          35m   10.1.254.73  microk8s-vm  <none>           <none>
ubuntu@microk8s-vm:~$
```

SUMMARY

This lab introduced Kubernetes as a container orchestration platform through hands-on use of a single-node cluster created with MicroK8s. The exercise demonstrated the roles of the control plane and worker node, the use of namespaces and add-ons, and how Kubernetes manages resources. An NGINX web server was deployed, tested, and scaled to multiple replicas to show workload scheduling and scalability. The lab concluded with cluster cleanup, reinforcing best practices for managing Kubernetes environments.